W-S-865c
SEPTEMBER 3, 1959

SUPERSEDING
Int. Fed. Spec. W-S-00865b (Navy-Docks)
May 22, 1957 and
Fed. Spec. W-S-865
July 27, 1950

FEDERAL SPECIFICATION

SWITCH, BOX, (ENCLOSED), SURFACE-MOUNTED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers surface-mounted, air-break, box or enclosed switches for ratings through 600 volts, 1200 amperes, and 50 horsepower.

1.2 Classification.

1.2.1 *Physical.*—The switches shall be of the following types, classes, designs, and boxes, as specified:

Type HD.—Heavy duty, single throw.

Type NDS.—Normal duty, single throw.

Type NDD. — Normal duty, double throw (transfer).

Type LD.—Light duty, single throw.

Class 0.—Not fused.

Class 1.—With one fuse.

Class 2.—With two fuses.

Class 3.—With three fuses.

Class 4.—With four fuses.

Design 2—2 blades, 2 poles.

Design 2SN — 2 blades, solid neutral, 3 poles.

Design 3—3 blades, 3 poles.

Design 3SN — 3 blades, solid neutral, 4 poles.

Design 4—4 blades, 4 poles.

Design 4SN — 4 blades, solid neutral, 5 poles.

Box 1.—General Purpose.

Box 3R.—Rain-tight.

Box 4.—Water-tight.

Box 12.—Industrial use.

1.2.2 *Voltage rating.* — The switch shall have one of the voltage ratings shown in table I, as specified.

1.2.3 *Ampere rating.* — The switch shall have one of the ampere ratings shown in table II, as specified. When so specified, the switch shall also be horsepower rated. (See 3.1.3 and 6.5.)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU-P-271 — Paper, Wrapping, Waterproofed Kraft.

PPP-B-585—Boxes, Wood, Wirebound.

PPP-B-591—Boxes, Fiberboard, Wood-Cleated.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

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PPP–B–621—Boxes, Wood, Nailed and Lock Corner.

PPP–B–636—Boxes, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking For Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–P–116—Preservation, Methods of.

MIL–B–138—Boxes, Wood, Fiberboard-Lined for Overseas Shipment (for weight of contents not exceeding 500 pounds).

MIL–A–140 — Adhesive, Water-Resistant, Waterproof Barrier Material.

Military Standards:

MIL–STD–129—Marking For Shipment And Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following publications form a part of this specification:

National Electrical Manufacturers' Association (NEMA):

KS–1—Enclosed Switches (1957).

(Copies may be obtained from the National Electrical Manufacturers' Association, 155 East 44th Street, New York 17, New York.)

Underwriters' Laboratories, Inc. Standards (UL):

UL–98—Enclosed Switches (Issue in effect on date of invitation for bids).

(Copies may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N. Y.; 207 East Ohio Street, Chicago 11, Ill.; or, 1655 Scott Lane, Santa Clara, Calif.)

3. REQUIREMENTS

3.1 General.

3.1.1 *Design.*—Each switch shall be simple in construction and reliable in operation. The switch mechanism shall be operable manually by means of a handle external to the box. All current carrying parts of the switch shall be enclosed.

3.1.2 *Performance.* — The switches shall conform to the requirements for heating, overload, endurance, and dielectric strength of UL–98, Standard for Enclosed Switches. Switches rated in horsepower shall also withstand the test for overload requirements of NEMA KS–1, Standard for Enclosed Switches. Type HD switches shall also withstand the endurance and the overload tests of the NEMA Standard.

3.1.3 *Horsepower rated switches.*—Horsepower rated switches shall have horsepower ratings as shown in Underwriters' Laboratories, Inc., Standard UL–98. Unless otherwise specified, fused switches shall be of the dual horsepower rated type wherever dual ratings are indicated in the UL Standard No. 98. Unless otherwise specified nonfusible switches shall have a rating equal to or above the maximum ratings shown in columns headed "Time delay fuse" of table III.

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3.2 Fire and casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency proof that the equipment he proposes to supply under this specification conforms to the requirements of Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories, Inc., may be accepted as evidence that the equipment conforms to this requirement.

3.2.2 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the contracting agency that his equipment conforms to the published standards including methods of tests of the Underwriters' Laboratories, Inc.

3.3 Switch box.—The switch box shall be of ample strength and rigidity to retain its shape under all conditions of use for which it is designed.

3.3.1 *Box 1—general purpose.*—General-purpose boxes shall be designed to prevent accidental contact with the enclosed switch and shall be suitable for general-purpose applications indoors where it is not exposed to unusual service conditions.

3.3.2 *Box 3R—rain-tight.* — Rain-tight boxes shall be constructed to prevent entrance of water above the level of the lowest live part of the enclosed switch when tested in accordance with 4.5.1, and to prevent entrance of drops of liquid or solid particles when striking the box at any angle up to 15 degrees from the downward vertical.

3.3.3 *Box 4—water-tight.* — Water-tight boxes shall be constructed to prevent entrance of water when tested in accordance with 4.5.2.

3.3.4 *Box 12—industrial use.*—Industrial-use boxes shall be constructed to exclude materials such as dust, lint, fibers and flyings, and oil and coolant seepage.

3.4 Switch-box cover interlocks.—Type HD switches shall have a dual interlock. Types NDS and NDD switches shall have a single interlock, unless a dual interlock is specified. Interlocks are not required on type LD switches.

3.4.1 *Single interlock.*—Single interlocks shall be designed to have the switch-box cover interlock with the switch mechanism so that the cover cannot normally be opened until the switch contacts are in the open, or "off" position; however, it shall be possible to close the switch contacts while the cover is open.

3.4.2 *Dual interlock.*—Dual interlocks shall be designed to have the switch-box cover interlock with the switch mechanism so that the cover cannot normally be opened until the switch contacts are in the open, or "off", position, and the switch contacts cannot normally be closed until the switch box cover is closed. However, this design of interlock shall be provided with a suitable means of interlock release so that the switch contacts may be placed in the closed, or "on", position while the switch box cover is opened.

3.5 Operating mechanism.—Unless otherwise specified, the operating handle may be either front or side operated. When moved to its "off" position, the handle shall positively operate the contacts to their "off" position without requiring spring action. For types HD, NDS, and NDD switches, rated 30 through 600 amperes, the operation of the contacts shall not be capable of being restrained by the operating handle after the opening or closing action of the contacts has started. The switching mechanism for types HD, NDS, and NDD switches shall operate the circuit making contacts in a quick-make, quick-break manner. Type LD switches, rated at 60 amperes and over, shall provide quick-break operation.

3.6 Fusing. — Unless otherwise specified (see 6.4), fusible switches shall have an accommodation for a cartridge-type fuse in each pole except solid-neutral and switching neutral poles. Each fused pole shall accom-

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modate a single fuse having a rating corresponding to the switch ampere rating for all switches rated at 600 amperes or less. Unless otherwise specified, fusible NDD switches shall be fused at both ends of the interruption.

3.7 Knockouts and conduit hubs.—Unless otherwise specified, conduit knockouts or hubs in the switch box shall be in accordance with the manufacturer's standard practice.

3.8 Locking. — When specified, switches shall have provisions for locking in the "on" position. All switches shall have provisions for locking in the "off" position.

3.9 Surface condition.—Box switches shall have no rough edges or burrs that may be injurious to persons operating the switches. The box shall have painted finish over a surface previously conditioned by a phosphate coating or equivalent process. Unless otherwise specified, the manufacturer's standard color for exposed visible parts of the box will be acceptable.

3.10 Identification marking.—The switches shall be marked in accordance with UL Standard No. 98, for enclosed switches.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein, the supplier is responsible for performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government for the time specified in the contract or order.

4.2 Sampling.—For purposes of sampling, a lot shall consist of all box switches of like characteristics and ratings offered for delivery at one time. Samples for inspection and tests shall be taken at random in an amount equal to at least 1 percent of each lot, but not less than 2 switches nor more than 10 switches. Samples selected shall be subjected to each of the examinations and tests specified herein. Failure to conform to the specification requirements for any inspection or test shall be counted as a defect and the lot shall be rejected.

4.3 Examination.—Each sample selected shall be given a general examination to determine conformance with this specification.

4.4 Performance tests.—Sample switches shall be tested for heating, overload, endurance and dielectric tests in accordance with the standards referenced in 3.1.2. The UL label will be accepted as evidence that the switch meets the requirements of the UL Standard No. 98 for Enclosed Switches. Test data showing that the switch has successfully met the requirements of this specification may be considered for acceptance in lieu of tests.

4.5 Box tests.

4.5.1 *Box 3R—rain-tight.*—A complete box with conduit connections (without pipe-thread compounds) shall be mounted as in actual service and a water spray, adjusted to be approximately the equivalent of a beating rain, shall be applied from the top and sides for an hour. The box shall be examined to determine compliance with 3.3.2.

4.5.2 *Box 4—water-tight.*—The box shall be subjected to a stream of water from a 1-inch nozzle delivering not less than 65 gallons per minute. The water shall be directed at probable weak spots on the box from a distance of 10 feet for a period of five minutes. After completion of the test, the box shall be examined to determine compliance with 3.3.3.

4.6 Rejected lots. — When the supplier elects to resubmit a rejected lot, the supplier shall first inspect the rejected lot, repair or remove all defectives, and then present the

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units as a resubmitted lot. The Government will elect to inspect for all types of defects or for only the defects that caused the lot to be rejected.

4.7 Inspection of preparation for delivery.—The preservation, packaging, packing, and marking of the switches shall be examined and tested to determine compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing used are defined in Federal Standard No. 102.

5.1 Preservation and packaging.—The preservation and packaging shall be level A or C, as specified.

5.1.1 *Level A.*

5.1.1.1 *Switches weighing not more than 70 pounds.*—Each switch weighing not more than 70 pounds shall be preserved and packaged in accordance with Military Specification MIL–P–116, method IA–2 or IA–15.

5.1.1.2 Switches weighing over 70 pounds.—Each switch weighing over 70 pounds shall be wrapped in waterproofed paper conforming to Federal Specification UU–P–271, class E–1 or E–2, and all laps and seams sealed with water-resistant adhesive conforming to Military Specification MIL–A–140. Projections of the switch shall be cushioned to prevent rupturing the wrapper.

5.1.2 *Level C.*—The switches shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing.—The packing shall be level A, B, or C, as specified.

5.2.1 *Level A.* — The switches shall be packed in close-fitting boxes conforming to Federal Specification PPP–B–621 (class 2), PPP–B–601 (overseas type), PPP–B–585 (class 3), PPP–B–591 (overseas type), PPP–B–636 (class 2), or Military Specification MIL–B–138. The contents shall be cushioned and braced to prevent movement within the boxes. The gross weight shall not exceed approximately 200 pounds for wood or wood cleated boxes, except when the weight of an individual switch exceeds 200 pounds. In no case shall the weight exceed the limitations of the specification for fiber boxes. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be flat, steel, type I, class B.

5.2.2 *Level B.* — The switches shall be packed in close-fitting boxes conforming to Federal Specification PPP–B–621 (class 1), PPP–B–601 (domestic type), PPP–B–585 (class 1), PPP–B–591 (domestic type), or PPP–B–636 (class 1). The contents shall be cushioned and braced to prevent movement within the boxes.

5.2.3 *Level C.* — The switches shall be packed or prepared for shipment in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates.

5.3 Marking.

5.3.1 *Nonmilitary agencies.*

5.3.1.1 *Interior packages.*—Each package or carton shall be plainly marked with the item name, specification number, type, class, design, box, voltage, ampere and horsepower rating, if any, and the name of the manufacturer.

5.3.1.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be plainly marked with the item name, specification number, type, class, design, box, voltage, ampere and horsepower rating, if any, the quantity contained therein, the name of the contractor, and the number of the contract or order. In addition to any special marking

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required by the contract or order, shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—Interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use.—These switches are intended for use where switch requirements are that the contacts interrupt the circuit in an air medium, have their current carrying parts enclosed in metal cases, and which are manually operable by means of an external handle.

6.1.1 *Type HD.*—Type HD switches are for application where safety, performance and continuity of service are emphasized or where special types of enclosures are required.

6.1.2 *Type NDS.*—Type NDS switches are for application where safety is essential and reasonable performance and continuity of service is required. These switches will normally be used on most permanent installations.

6.1.3 *Type NDD.*—Type NDD switches are transfer switches.

6.1.4 *Type LD.*—Type LD switches are intended for application where service is relatively light and for temporary installations.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, design, box, voltage, and ampere rating. (See 1.2.1, 1.2.2 and 1.2.3).
(*c*) When horsepower ratings are required (see 1.2.3 and 3.1.3). When single horsepower rating is desired for fusible switches, or, when other than maximum rating is desired for non-fusible switches, it should be so stated (see 1.2.3 and 6.5).
(*d*) When dual interlocks are required for types NDS and NDD (see 3.4).
(*e*) When specific location of switch handle is required (see 3.5).
(*f*) When plug type fuse switches are required (see 3.6 and 6.4).
(*g*) When type NDD switches are to be fused on one end only (see 3.6).
(*h*) The number, size, and location of conduit knockouts or conduit hubs, when desired (see 3.7).
(*i*) When provisions for locking switches in the "on" position are required (see 3.8 and 6.6).
(*j*) When specific color is required (see 3.9).
(*k*) Any special inspection instructions and lengths of time inspection records must be kept available (see 4.1).
(*l*) Levels of preservation, packaging, packing, and marking required (see 5.1, 5.2, and 5.3).

6.3 Enclosures. — Type HD switches are normally furnished with general purpose (box 1), water-tight (box 4), or industrial (box 12) boxes. Types NDS, NDD, and LD switches are normally furnished in general purpose (box 1), or rain-tight (box 3R) boxes.

6.4 Fuse. — Plug-type fuse switches are only available in the 30 ampere, 125 volt rating.

6.5 Horsepower rating.—The horsepower ratings of fusible switches available under this specification are as shown in Table III. The horsepower ratings of non-fusible switches available under this specification are as shown in columns headed "Time-delay fuse" of Table III.

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6.6 Locking operating mechanism.—Disconnecting switches have provisions for locking the contacts in the "on" position. Current interrupting switches may be provided with means for locking the contacts in the "on" position when this feature is required.

6.7 Precautionary notes.

(*a*) Fusible switches intended for continuous-duty loads other than motor loads must be selected to accommodate fuses 25 percent higher in rating than the load current.

(*b*) Switches marked "for disconnecting (isolating) use" are not intended to be used for opening circuits under load, and should not be selected for such service.

(*c*) Switches rated at more than 600 amperes at more than 250 volts are for disconnecting use only.

6.8 Federal Specifications do not include all types, classes, etc. of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.9 Transportation description.—Transportation description applicable to this item is:

Switches, not otherwise indexed by name, electric.

Carload minimum weight 30,000 pounds.

Motor volume minimum weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

TABLE I.—*Voltage ratings*

Design	Class	Voltage rating	
		DC*	AC
2	0, 1, or 2	125 250	120 240
	0 or 2	125/250 600	120/240 480 600
2SN	0 or 2	125/250	120 120/240 240
3	0, 2, or 3	-------	120 240
	0 or 2	125/250	120/240
	0 or 3	-------	480 600
3SN	0 or 3	-------	240 480 600
4	0 or 4	-------	120
	0, 3, or 4	-------	240 480 600
4SN	0 or 4	-------	120/240

* Direct-current rated switches are suitable for alternating-current service of equal or lower voltage rating.

TABLE II.—*Ampere ratings*

Type	Ampere ratings
HD	30, 60, 100, 200, 400, and 600
NDS	30, 60, 100, 200, 400, 600, 800, and 1200
NDD	30, 60, 100, 200, 400, and 600
LD	30, 60, 100, and 200

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TABLE III.—*Horsepower ratings*

Ampere rating	Voltage rating	Two-pole single-phase		Two-pole DC		Three-pole, Three-phase		Four-pole, Two-phase	
		Standard fuse	Time-delay fuse	Standard fuse	Time-delay fuse	Standard fuse	Time-delay fuse	Standard fuse	Time-delay fuse
30	120 AC	¾	2	--	--	1½	3	2	3
60		1½	3	--	--	3	7½	3	10
30	125 DC	--	--	2	3	--	--	--	--
60		--	--	5	5*	--	--	--	--
30	240 AC	1½	3	--	--	3	7½	3	10
60		3	10	--	--	7½	15	7½	20
100		7½	15	--	--	15	30	15	30
200		15	15*	--	--	25	50	30	50
400		--	--	--	--	50	50*	50	50*
30	250 DC	--	--	5	5*	--	--	--	--
60		--	--	10	10*	--	--	--	--
100		--	--	20	20*	--	--	--	--
200		--	--	40	40*	--	--	--	--
400		--	--	50	50*				
30	480 AC	3	7½	--	--	5	15	7½	20
60		5	20	--	--	15	30	15	40
100		10	30	--	--	25	50	25	50
200		25	50	--	--	50	50*	50	50*
30	600 AC	3	10	--	--	7½	20	10	25
60		10	25	--	--	15	50	20	50
100		15	40	--	--	30	50	30	50
200		30	50	--	--	50	50*	50	50*
30	600 DC	--	--	10	15	--	--	--	--
60		--	--	25	30	--	--	--	--
100		--	--	40	50	--	--	--	--
200		--	--	50	50*	--	--	--	--

* Applies to non-fusible switches only.

MILITARY CUSTODIANS:
Army—Corps of Engineers
Navy—Bureau of Yards & Docks
Air Force.

PREPARING ACTIVITY:
Navy—Bureau of Yards and Docks

Copies of this specification may be purchased for 5 cents each.

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